

Haptic-aware Interaction: Design and Evaluation

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ABSTRACT

The emerging haptic technology has introduced new media perceptions and also increased the immersive experiences of end-users. To date, novel haptic-audio-visual environments and their Quality of Experience (QoE) assessments are still challenging issues. In this work, we investigate the haptic-visual interaction QoE in virtual as well as real-world environments. First, we establish a haptic-visual interaction platform based on a balance ball Virtual Reality (VR) game scene and a haptic-visual interaction platform with data-glove-assisted remote control. Second, we conduct subjective tests to qualitatively and quantitatively analyze the impacts of system-related, user-related and task-related factors on QoE evaluation. Third, we propose learning-based QoE models to effectively evaluate the user-perceived QoE in haptic-visual interaction. In the future work, we aim to focus on the improvement of the two established platforms, with the addition of audio-related influencing factors and more haptic feedback, and optimizing the proposed QoE model for further improvement of haptic-audio-visual interaction applications.

CCS CONCEPTS

• **Human-centered Computing** → **Human computer interaction (HCI)**.

KEYWORDS

multimedia interaction, haptic, quality of experience, immersion

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1 INTRODUCTION

With the rapid development of multimedia communications, massive multimedia applications have been able to satisfy users' audio-visual demands[21]. People start to pursue more immersive and interactive experiences. They are paying attention to touch sensory, which refers to as haptic technology. Haptic data consists of kinesthetic and tactile information [10]. Kinesthetic information includes all body movement data, such as position, force, velocity, torque, and

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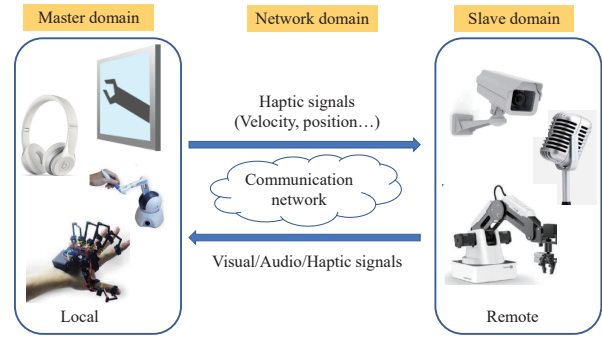


Figure 1: Typical HAV interaction system.

angular velocity. Tactile information refers to an object's surface properties, including shape, surface texture, and temperature. The integration of haptic information in audio-visual multimedia interaction applications can enhance the user's immersive experiences and improve operation accuracy [1]. Haptic-Audio-Visual (HAV) interaction systems will be widely used in industrial automation, autonomous driving, smart grid, gaming, health and education.

A typical HAV interaction system is shown in Fig. 1. The system consists of a master domain, a slave domain and a network domain [10] [15]. Specifically, the master domain to send commands includes human operator and human-machine interfaces, such as haptic devices, video displays, and headphones. The slave domain to execute commands contains a controlled robot arm equipped with multiple haptic sensors, a video camera and a microphone. The network domain provides efficient, bi-directional exchange of haptic, audio and visual information. At the master domain, a human operator interacts with the haptic interface and transmits haptic signals over the communication channel to the slave domain. The slave domain responds with the audio, visual and haptic feedback of the remote environment. An interaction system that integrates audio, visual and haptic signals can provide a much more immersive experience and effectively improve the user's interaction effects.

With the popularity of the use of haptics, researchers are paying more attention to user Quality of Experience (QoE) evaluation in the human-in-the-loop HAV interaction system. QoE plays a crucial role in the design and optimization of signal processing, compression, and transmission [20]. To date, QoE-related technologies in audio-visual interaction have been relatively mature, including subjective evaluation methods and objective QoE assessment models. These studies adequately meet the demands of audio-visual interaction. However, these studies are less suitable for HAV interaction. The characteristics of haptic signals are different from those of audio and visual signals. In particular, kinesthetic interactions are delay-sensitive [8]. On the other hand, haptic signals

are bidirectionally exchanged over the network. It closes a global control loop between operator and teleoperator. QoE must extend from audio-visual interaction to HAV interaction.

So far, only very few studies for QoE evaluation for HAV interaction are available. Existing studies qualitatively analyzed the some influencing factors of haptic interaction QoE, such as haptic transmission delay, delay jitter, intra-stream synchronization and haptic data reduction. However, in the human perception system, there exist inherent interactions among auditory, haptic, and visual sensory subsystems, which comprehensively determine humans experience [7], [14]. In addition to haptic-related influencing factors, audio-related and visual-related factors also have an impact on HAV interactions. On the other hand, haptic QoE assessment models have been proposed based on mathematical derivations, but these models suffer from feasibility and accuracy issues as there is no comprehensive model that can quantify the multidimensionality and large individual variability [10]. Moreover, the existing studies are mainly based on virtual environments, which are less suitable for HAV teleoperation in a real-world environment.

Aiming at investigating the QoE of haptic-aware interaction, we create the haptic-visual interaction platforms in virtual and real-world environments respectively, analyze the key QoE influencing factors, and present the learning-based QoE evaluation of haptic-visual interaction.

The rest of this paper is organized as follows. Section 2 gives the related work. Section 3 presents the established haptic-aware interaction platforms. Section 4 describes the proposed QoE evaluation methodology and the preliminary results. Section 5 concludes the paper with a summary of the presented research.

2 RELATED WORK

This section contains a brief view of related research regarding two perspectives, including haptic-aware interaction systems and the QoE assessment for haptic-aware interaction applications.

2.1 Haptic-aware Interaction Design

Many researchers have attempted to build haptic-aware interaction systems. Sun *et al.* [11] presented a haptic-feedback ring-enabled glove-based human-machine interface for advanced interaction and collaborative operation in robotic manipulation and VR areas. Dong *et al.* [3] present an immersive virtual reality training system, in which users can interact with the equipment in the virtual environment by moving or touching physical models. Xu *et al.* [16] proposed a haptic-visual aid teleoperation system that deal with large time delay and obstacle avoidance. Yang *et al.* [17] combined human intelligence with the robot's capability to enhance the robot intelligence in the teleoperation tasks. Yu *et al.* [18] proposed a hybrid framework using visual and force information for human-robot co-carrying tasks. The framework enabled the robot to proactively follow its human partner and reduce control effort. The design of haptic-aware interaction in both virtual reality and real-world environments has received significant attention, which has driven research on the QoE of haptic-aware interaction.

2.2 Haptic-aware Interaction Evaluation

Compared with QoE-related technologies in audio-visual interaction, QoE in haptic-visual interaction is a relatively young research direction. The studies mainly focus on influencing factor analysis and objective QoE evaluation metrics. Shi *et al.* [9] simulated random packet dropouts and communication latency in the visual modality and investigated the effects on the temporal discrimination of haptic-visual collisions. Chaudhari *et al.* [2] investigated the impact of haptic data reduction on QoE and proposed a framework for objective quality evaluation of haptic compression schemes. Zeng *et al.* [19] investigated the influence of inter-stream synchronization errors among haptic media, sound, and video on QoE assessment in a collaborative virtual environment. Tatematsu [4] investigated the influence of network delay, jitter and inter-stream synchronization error on the QoE of a teleoperation system with stereoscopic video and haptic media. Hamam *et al.* [5] proposed a taxonomy for the evaluation of haptic-aware systems' QoE based on a virtual reality game called the Balance Ball game. The taxonomy was divided into Quality of Service (QoS) and User Experience (UX) groups. They investigated the impact of media synchronization, fatigue, haptic rendering, degree of immersion and user intuitiveness. Then, they provided a mathematical model based on a weighted linear combination between QoS and UX parameters. A fuzzy logic inference system was further implemented to quantitatively measure the QoE of haptic virtual environments [6]. Tasaka [12] [13] proposed Bayesian structural equation models for assessing multidimensional QoE by introducing audiovisual quality, haptic quality, and user experience quality. These research efforts only focus on haptic-visual interaction in virtual environments.

Given the above discussion, QoE assessments for haptic-visual interaction are still a challenging problem. In order to provide a much more immersive experience and improve the user's interaction effects, it is strongly desirable to investigate the influence factors of haptic-visual interaction QoE in virtual and real-world environments and develop a QoE objective evaluation metric.

3 PLATFORM CONSTRUCTION

In order to analyze the QoE influencing factors in haptic-aware interaction, we establish two interaction platforms for teleoperation in virtual and real environments, respectively. The two interaction platforms both includes a master domain, a slave domain and a bi-directional communication channel.

3.1 Haptic-aware Virtual Interaction

In our first stage of the research, we mainly analyze the haptic-aware interaction based on a balance ball VR game scene. The haptic-aware interaction system architecture is shown in Fig. 2.

The master domain includes a haptic device (Geomagic Touch) with a sampling rate of 1000 Hz, a computer, a screen-based eye tracker (TobiiPro Spectrum 150) and data analysis software (Tobii ProLab). The operator watches the eye tracker screen and operates the haptic device to complete the virtual interaction tasks. Haptic data, including position coordinates, direction, grip angle, line speed, angular speed and grip angular speed, are captured and sent to the slave domain over the communication channel. Meanwhile,

the operator's eye movement information, including the gaze points, gaze directions, and fixation types, is recorded.

The slave domain includes a virtual interaction platform and a camera. The virtual interaction interface is rendered using the received haptic information. The operator controls a virtual gripper to grip the target object and move it to the target area by manipulating the Geomagic Touch. The camera captures the virtual interaction interface. Then the visual information is sent to the master domain over the communication channel.

A communication network based on the UDP/IP protocol is employed to transmit the haptic and visual information in real time.

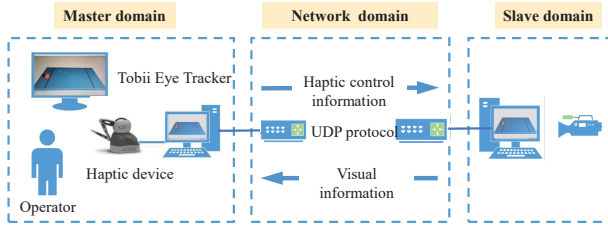


Figure 2: Haptic-visual aware interaction platform based on a balance ball VR game scene

3.2 Haptic-Visual Teleoperation

In the second phase of the study, in order to further explore the influencing factors of haptic-aware interaction QoE in a real-world environment, we create a haptic-visual interaction platform with data-glove-assisted remote control, as shown in Fig. 3.

The master domain consists of two computers, a force feedback glove as a haptic interface device and a display. The computers are connected to each other via a router. The glove is connected to one of the computers by Wi-Fi, and the other computer is used to display visual information. An operator operates the robot arm with the force feedback glove while watching visual information, as shown in Fig. 4. The operator uses the thumb to control the opening and closing of the robotic arm's jaws, the mid-finger to control the rotation of the robotic arm base, and the forefinger and ring finger to control the up and down movement of the robotic arm's large and small arms, respectively. The acceleration and position information for five fingers in the x , y , and z axes, are transmitted to the slave domain. Meanwhile, the master domain receives the visual information from the slave domain to monitor the robotic arm's command execution and facilitate control command adjustment.

The slave domain consists of two computers, an industrial robotic arm and a camera. The computers are connected to each other via a router. The robot arm and camera are connected to the two computers, respectively. The slave domain receives the acceleration and position information. Based on this information, different control commands are generated and sent to the robotic arm. The robotic arm executes control commands and completes tasks. The visual information of the robotic arm motion captured by the camera is transmitted to the master domain.

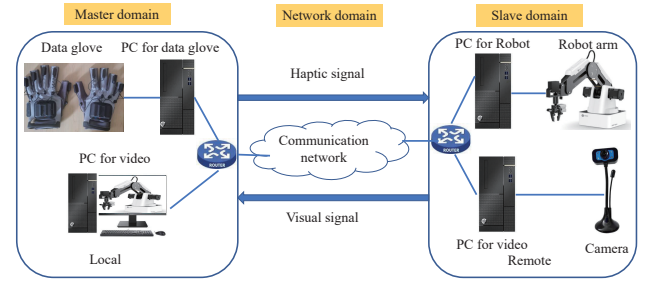


Figure 3: Haptic-visual interaction platform with data-glove-assisted remote control.

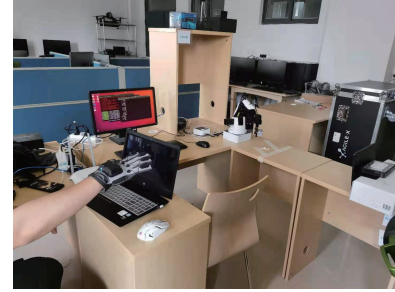


Figure 4: Haptic-visual interaction demo. An operator operates the robot with the force feedback glove while watching visual information.

4 QOE EVALUATION

In this section, we discuss the work that has been done so far toward haptic-aware interaction QoE evaluation.

The taxonomy in ([5]) groups QoE into QoS and UX. We further divide the influencing factors into three categories: task-related, system-related, and user-related, as shown in Table 1. Audio, video or haptic transmission quality is highly correlated with the network environment and can be measured using transmission delay, delay jitter, packet loss and transmission error. The physical environment includes light, noise, and so on.

Table 1: The QoE influence factors in haptic-aware interaction

Category	Master domain	Network domain	Slave domain
System-related	Video quality	Video transmission quality	Video quality
System-related	Audio quality	Audio transmission quality	Audio quality
System-related	Haptic quality	Haptic transmission quality	Haptic quality
System-related	Physical environment	Network environment	Physical environment
Task-related	/	/	Task complexity
Task-related	Task result	/	/
Task-related	User operation time	/	/
User-related	Emotion states	/	/
User-related	Perception states	/	/
User-related	Physiological states	/	/
User-related	Prior knowledge	/	/

First, we analyze the QoE-influencing factors based on two interaction platforms and select the key factors. Second, we design subjective evaluation experiments and establish QoE subjective evaluation databases. Third, we analyze the relationship between the influencing factors and QoE and propose learning-based QoE evaluation models.

Based on the balance ball VR game scene, six influencing factors are selected, including video transmission delay, haptic data loss, haptic perceptual deadband parameter, task complexity, task result, and user operation time. We construct 81 haptic interaction tasks and collect the subjective data. We calculate the Pearson Linear Correlation Coefficient (PLCC), Spearman Rank-order Correlation Coefficient (SRCC) and Kendall Rank-order Correlation Coefficient (KRCC) between each influencing factor and the Mean Opinion Score (MOS) values. The results are shown in Fig. 5.

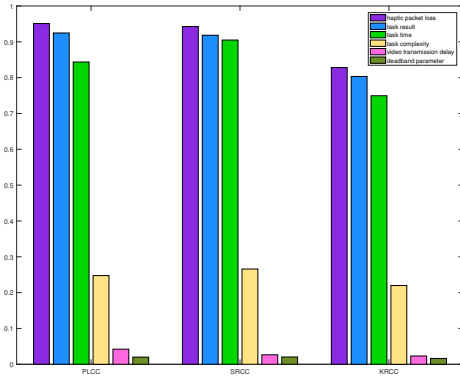


Figure 5: Correlation between each influencing factor and QoE in the balance ball VR game scene.

We utilize machine learning algorithms to construct a QoE evaluation model and evaluate its performance. Four machine learning algorithms, including Linear Regression (LR), Support Vector Regression (SVR), k-Nearest Neighbor (kNN) and decision tree regression (DTR), are used to explore the relationship between key factors and MOS. PLCC, SRCC, KRCC and Mean Squared Error (MSE) between the predicted and authentic values are utilized to evaluate the model performance, as shown in Table 2.

Table 2: Performance evaluation of the different models for haptic-aware virtual interaction

Model	PLCC	SRCC	KRCC	MSE
LR	0.9796	0.9398	0.8654	0.0430
DTR	0.9775	0.9511	0.8953	0.0486
kNN	0.9698	0.9295	0.8467	0.0655
SVR	0.9775	0.9511	0.8953	0.0486

Based on the haptic-visual interaction platform with data-glove-assisted remote control, we analyze the impact of video transmission delay, video packet loss, haptic data loss, the spatial resolution

of video, task complexity and task result on QoE. The subjective database contains a total of 243 records. The PLCC, SRCC and KRCC between each influencing factor and MOS values are shown in Fig. 6.

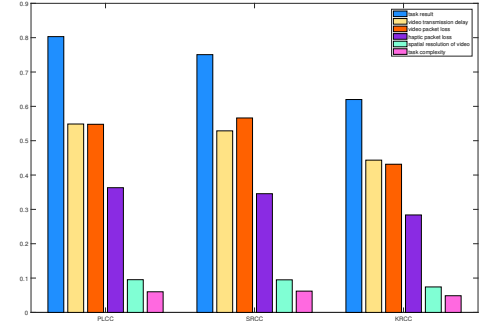


Figure 6: Correlation between each influencing factor and QoE based on data-glove-assisted remote control.

LR, SVR, kNN and DTR are used to explore the relationship between key factors and MOS. PLCC, SRCC, KRCC and MSE between the predicted and authentic values are shown in Table 3. DTR outperforms other approaches in terms of PLCC, SRCC, KRCC and MSE. The results in Tables 1 and 2 indicate that the DTR-based model can effectively evaluate the user-perceived QoE in haptic-visual interaction.

Table 3: Performance evaluation of the different models for data-glove-assisted remote control

Model	PLCC	SRCC	KRCC	MSE
LR	0.8826	0.8968	0.7330	0.2145
DTR	0.9316	0.9195	0.7928	0.1363
kNN	0.8410	0.8420	0.6554	0.2880
SVR	0.8958	0.8816	0.6892	0.3192

5 CONCLUSION AND FUTURE WORK

We have established the interaction platforms for teleoperation in virtual and real environments, respectively. Moreover, we have selected key QoE influencing factors and established subjective evaluation databases. Then we have preliminarily analyzed the relationship between these factors and QoE and proposed the DTR-based QoE model to effectively evaluate the user-perceived QoE. In the present study, we have not considered the effect of audio and haptic feedback from the slave domain on QoE. In the future work, we will optimize the two established platforms, provide audio feedback and haptic feedback from the slave domain, explore influencing factors of QoE more comprehensively, and construct a more general QoE evaluation model.

6 ACKNOWLEDGEMENTS

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